STAT5080 – HW7 Solutions

Md Zehen Alam & Michael Abalo

Date: 04/02/26

Problem 1:

An industrial engineer investigates how the **distance of an object from the eye** affects **focus time**. Four distances (4, 6, 8, 10 ft) are studied with five subjects acting as blocks to control individual variability.

Part (a): Name of the Design

Answer: This is a **Randomized Block Design (RBD)**, specifically, with one observation per treatment per block (s = 1). The **treatment factor** is *distance* (4 levels) and the **blocking factor** is *subject* (5 levels). The block size equals the number of treatments (k = a = 4), and each treatment appears exactly once in every block.

Part (b): How the Data Were Collected

Answer: The researcher identified five subjects as experimental units grouped into blocks to account for individual differences in focus ability. For each subject (block), the four distances (4, 6, 8, and 10 ft) were randomly assigned to the four trials, a separate, independent randomization within each subject. This restriction on randomization is the hallmark of a block design: it ensures that each treatment appears exactly once per block while variability due to subjects is isolated and removed from the error term. The randomized block design improves the precision of detecting distance effects by eliminating between-subject noise from the experimental error.

Model:

The standard RBD (no-replication, no interaction) model is:

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}, \quad i = 1, \dots, 4;\ j = 1, \dots, 5$$

where τ_i are fixed treatment effects (distance), β_j are fixed block effects (subject), and $\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2)$.

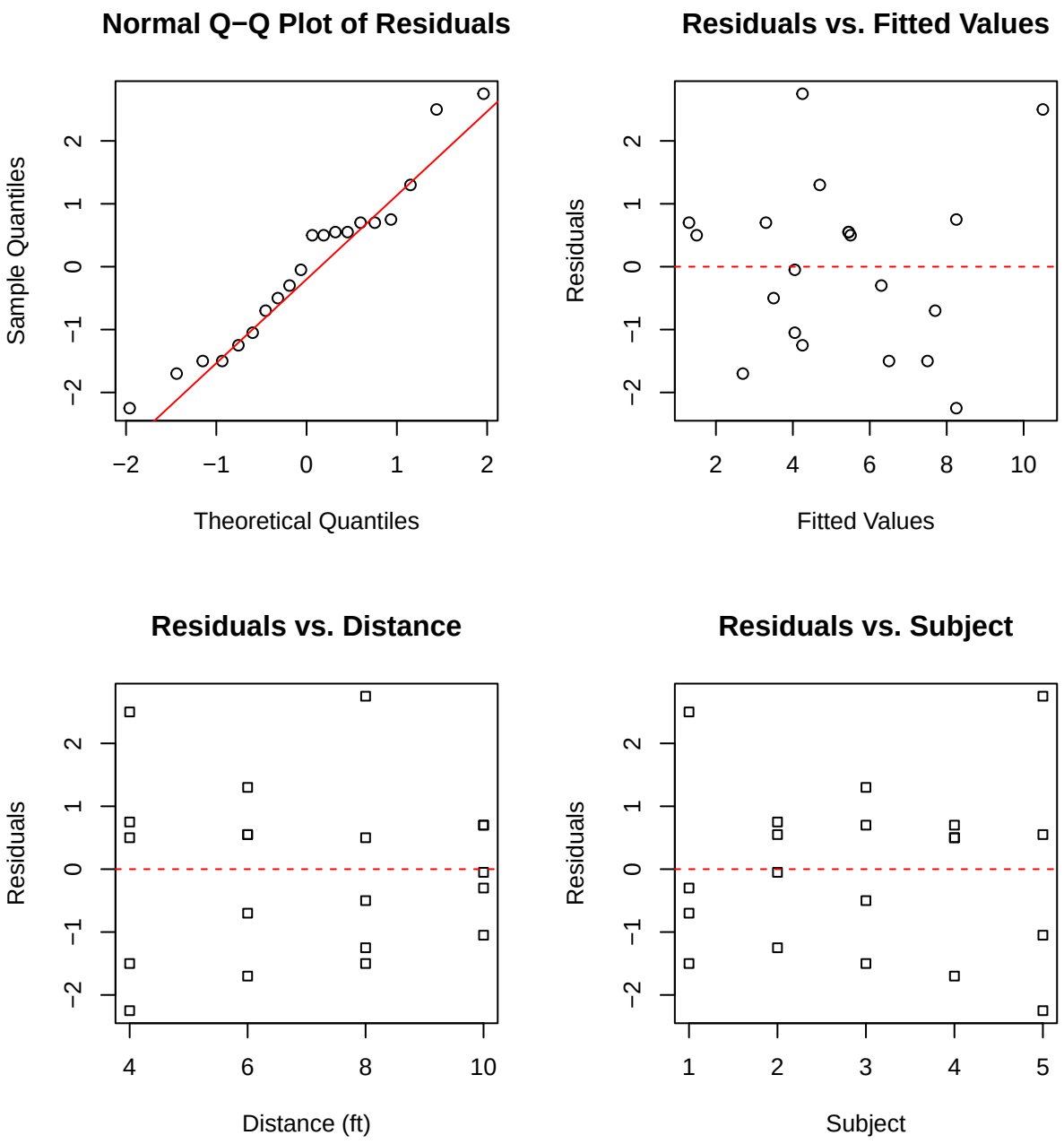
Part (c): Analysis and Conclusions

ANOVA

```
## Analysis of Variance Table
##
## Response: focus
##           Df Sum Sq Mean Sq F value   Pr(>F)
## distance   3  56.15 18.7167  6.5865 0.007012 **
## subject    4  51.50 12.8750  4.5308 0.018423 *
## Residuals 12  34.10  2.8417
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation: The F-test for **Distance** has a p-value of 0.007. Since this is less than $\alpha = 0.05$, we **reject H :** $=$ $=$ $=$ and conclude there is sufficient evidence that distance significantly affects mean eye focus time. The MSE (error mean square) is 2.8417, which estimates σ^2.

Assumption Checking

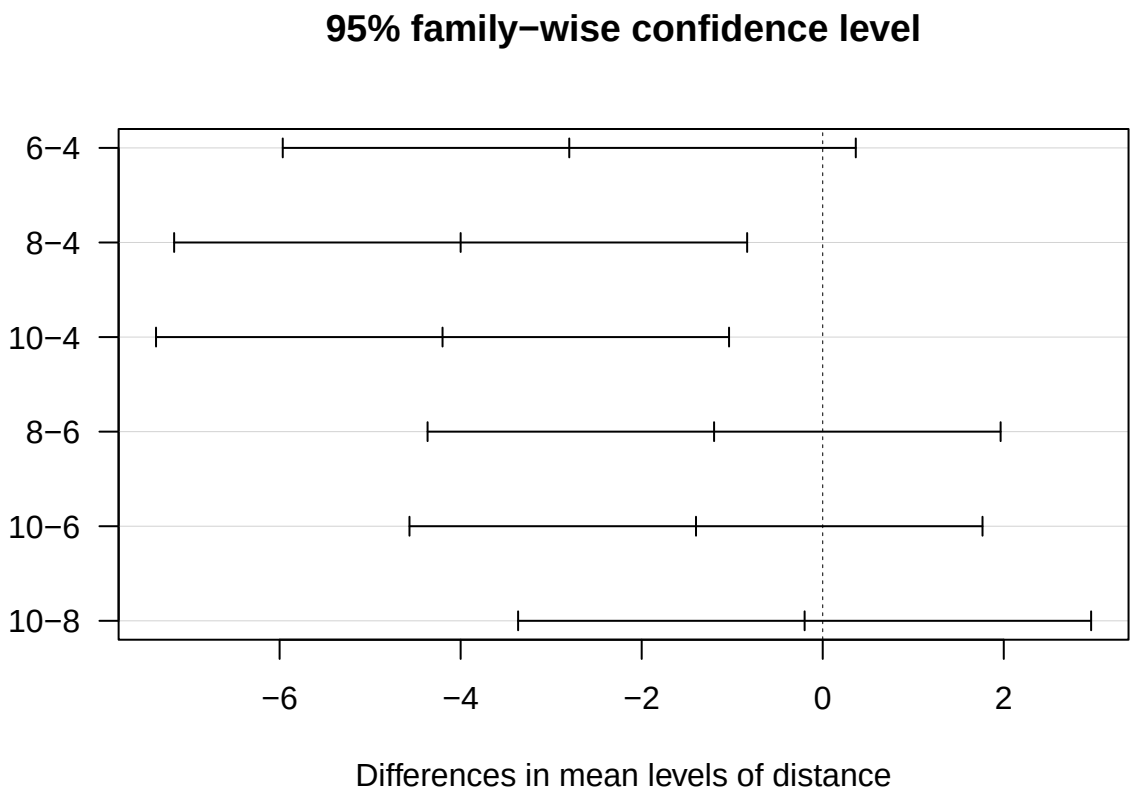


Interpretation of residual plots:

- **Q-Q plot:** Points follow the reference line reasonably well, suggesting no major departure from normality.
- **Residuals vs. Fitted:** No strong fan or curved pattern; the constant variance assumption appears tenable.
- **Residuals vs. Distance / Subject:** Residuals are roughly centered around zero across all levels; no evidence of systematic patterns.

Overall, the model assumptions appear to be adequately satisfied.

Treatment Means and Tukey Multiple Comparisons




```
##   4   6   8  10
## 8.0 5.2 4.0 3.8
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = focus ~ distance + subject, data = eyedata)
##
## $distance
##      diff       lwr        upr     p adj
## 6-4  -2.8 -5.965281  0.3652812 0.0895859
## 8-4  -4.0 -7.165281 -0.8347188 0.0127336
## 10-4 -4.2 -7.365281 -1.0347188 0.0091790
## 8-6  -1.2 -4.365281  1.9652812 0.6815580
## 10-6 -1.4 -4.565281  1.7652812 0.5724264
## 10-8 -0.2 -3.365281  2.9652812 0.9975253
```

The mean focus time at **4 ft**(8.0) is significantly longer than the focus times at **8 ft** (4.0) and **10 ft** (3.8). We know this because their respective confidence intervals ([-7.17, -0.83] and [-7.37, -1.03]) do not contain zero, and their p-values are well below 0.05.

The difference between **4 ft** and **6 ft** is not statistically significant at the standard =0.05 level, as the interval [-5.97, 0.37] includes zero (p-adj = 0.0896). There are no significant differences in focus times among the **6 ft, 8 ft, and 10 ft** distances, as all pairwise comparisons between these groups contain zero in their confidence intervals.

Overall Conclusion: Shorter distances (specifically 4 ft) significantly increase eye focus time compared to longer distances (8 ft and 10 ft). However, there is no statistically significant evidence of a difference in focus time once the object is 6 feet or further away.

Problem 2

Part (a): RCBD - 5 Treatments, 4 Blocks (1 observation each)

In a Randomized Complete Block Design, every treatment appears exactly once in every block. Within each block, treatments are randomly assigned to experimental units independently of other blocks.

```
## Randomization for RCBD (5 treatments, 4 blocks, s=1):
##         Unit_1 Unit_2 Unit_3 Unit_4 Unit_5
## Block_1      1      4      3      5      2
## Block_2      5      3      4      2      1
## Block_3      3      5      1      4      2
## Block_4      2      5      4      3      1
##
## Each row is a block; the numbers indicate which treatment (1-5) is assigned to each unit.
```

Each row (block) is an independent random permutation of the 5 treatments. Every treatment appears exactly once per block (k = a = 5), satisfying the RCBD requirement. Total observations = $4 \times 5 = 20$.

Part (b): General Complete Block Design — 5 Treatments, 4 Blocks (s = 2 each)

When $s = 2$, each treatment is observed **twice** per block, so the block size is $k = as = 10$. The randomization assigns two units per treatment within each block.

```
## Randomization for General RCBD (5 treatments, 4 blocks, s=2):
##         Unit_1 Unit_2 Unit_3 Unit_4 Unit_5 Unit_6 Unit_7 Unit_8 Unit_9 Unit_10
## Block_1      5      3      5      1      4      4      3      1      2       2
## Block_2      5      1      2      2      3      1      3      4      5       4
## Block_3      2      5      4      3      1      5      4      2      1       3
## Block_4      5      4      2      1      3      5      1      3      4       2
##
## Each row = block; each treatment (1-5) appears exactly twice per block.
```

Each block contains 10 experimental units, and within each block, the 10 slots are randomly assigned so that each of the 5 treatments occupies exactly 2 units. Independent randomization is performed within each block. Total observations = $4 \times 10 = 40$.

Part (c): Incomplete Block Design

In an incomplete block design (IBD), not all treatments appear in every block. The specified structure (Table A) assigns exactly 3 treatments to each of 9 blocks. Randomization means that within each block, we randomly assign the listed treatments to the available experimental units.

```
## Randomization within each block (3 units per block):
##
## Block      Treatments      Randomized Assignment
## ------------------------------------------------------------
## Block1     1 2 3           2 -> 1 -> 3
## Block2     4 5 6           6 -> 5 -> 4
## Block3     7 8 9           9 -> 7 -> 8
```

```
## Block4     1 4 7          4 -> 7 -> 1
## Block5     2 5 8          8 -> 2 -> 5
## Block6     3 6 9          3 -> 6 -> 9
## Block7     1 5 9          1 -> 9 -> 5
## Block8     2 6 7          7 -> 2 -> 6
## Block9     3 4 8          4 -> 3 -> 8
```

For each block, the treatments assigned to that block are randomly allocated to the 3 experimental units within the block (independent randomization per block). The "Randomized Assignment" column shows the random order in which units within each block receive treatments.

Part (d): Is the Design in (c) a BIBD?

No, this isn't a Balanced Incomplete Block Design (BIBD).

A BIBD requires:

1. Each treatment has an equal number of replications (r).
2. Each treatment occurs at most once per block.
3. Every pair of distinct treatments appears together in exactly λ blocks.

```
## Replications per treatment:
## 1 2 3 4 5 6 7 8 9
## 3 3 3 3 3 3 3 3 3
##
## All treatments replicated r = 3 times?
##  TRUE
##
## Co-occurrence counts (upper triangle):
##  [1] 1 1 1 1 0 1 1 1 0 1 0 1 1 1 1 1 1 0 1 0 1 0 1 1 1 1 0 1 1 0 1 0 1 1 1 1
##
## All pairs appear together lambda = 1 0 times?
##  FALSE
##
## BIBD identity check: lambda*(a-1) = 8  and r*(k-1) = 6
```

While all nine treatments are each replicated **3** times and every treatment occurs at most once per block, every pair of distinct treatments **does not** co-occur an equal number of times (some pairs co-occur **1** time, while others co-occur **0** times). Because a constant λ does not exist, the required BIBD identity $\lambda(a-1) = r(k-1)$ fails. If we attempt to evaluate it using a target λ of **1**, the left side yields $1 \times (9-1) = 8$, while the right side yields $3 \times (3-1) = 6$. Since $8 \neq 6$, this is **not** a valid Balanced Incomplete Block Design (BIBD).

Problem 3:

Seven hardwood concentrations (treatments) are tested across 7 days (blocks), with only 3 runs possible per day. This calls for an incomplete block design.

Part (a): Table A Format (Block–Treatment Listing)

```
## Table A: Blocks and Treatment Assignments
## ======================================
##  Block_Day Treatments
##          1    2, 4, 8
##          2   4, 6, 10
##          3   6, 8, 12
##          4  8, 10, 14
##          5  2, 10, 12
##          6  4, 12, 14
##          7   2, 6, 14
```

Part (b): Justification That This Is a BIBD

```
## Replications per concentration:
##  2  4  6  8 10 12 14
##  3  3  3  3  3  3  3
##
## Equal replication (r = 3 ): TRUE
##
## Co-occurrence values (should all be equal):
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
##
## lambda = 1 - all pairs appear together lambda times? TRUE
##
## BIBD parameters: a = 7 , b = 7 , k = 3 , r = 3 , lambda = 1
## BIBD identity lambda*(a-1) = 6 ; r*(k-1) = 6
```

Justification: This is a BIBD because: (1) each of the 7 hardwood concentrations is tested exactly $\mathbf{r = 3}$ times; (2) each concentration appears at most once per day (block); and (3) every pair of distinct concentrations is tested together in exactly $\mathbf{= 1}$ block. The BIBD identity $\lambda(a-1) = r(k-1)$ holds: $1 \times 6 = 3 \times 2 = 6$.

Part (c): How the Design Was Implemented and Data Collected

The pilot plant can produce only $\mathbf{k = 3}$ **runs per day**, so a complete block design (which would require all 7 concentrations per day) is infeasible. A BIBD with parameters $(a = 7, b = 7, k = 3, r = 3, \lambda = 1)$ was constructed so that every concentration is tested on 3 of the 7 days and every pair of concentrations is tested together on exactly 1 day, ensuring balanced pairwise comparisons.

For data collection, on each day (block), the **3 assigned concentrations were randomly ordered** within that day's production runs (independent randomization per block).

The paper's breaking strength was recorded for each run. Because each concentration is tested across different days, day-to-day variability is accounted for through the block structure rather than inflating the error.

Part (d): BIBD Analysis and Conclusions

Data Entry and ANOVA

```
## Analysis of Variance Table
##
## Response: strength
##           Df  Sum Sq Mean Sq F value   Pr(>F)
## day        6 1150.29 191.714  10.180 0.002223 **
```

```
## conc       6 1306.00 217.667  11.557 0.001445 **
## Residuals  8  150.67  18.833
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The F-test for **Hardwood Concentration** (entered after Day to obtain adjusted/Type I SS) yields a p-value of 0.0014. Since this is less than $\alpha = 0.05$, we **reject H** and conclude that hardwood concentration has a significant effect on paper strength.

```
## Least Squares Means (adjusted for blocks):
##  conc emmean  SE df lower.CL upper.CL
##  2       117 2.8  8      111      124
##  4       121 2.8  8      115      127
##  6       132 2.8  8      125      138
##  8       140 2.8  8      134      147
##  10      143 2.8  8      137      150
##  12      124 2.8  8      118      131
##  14      129 2.8  8      122      135
##
## Results are averaged over the levels of: day
## Confidence level used: 0.95
```

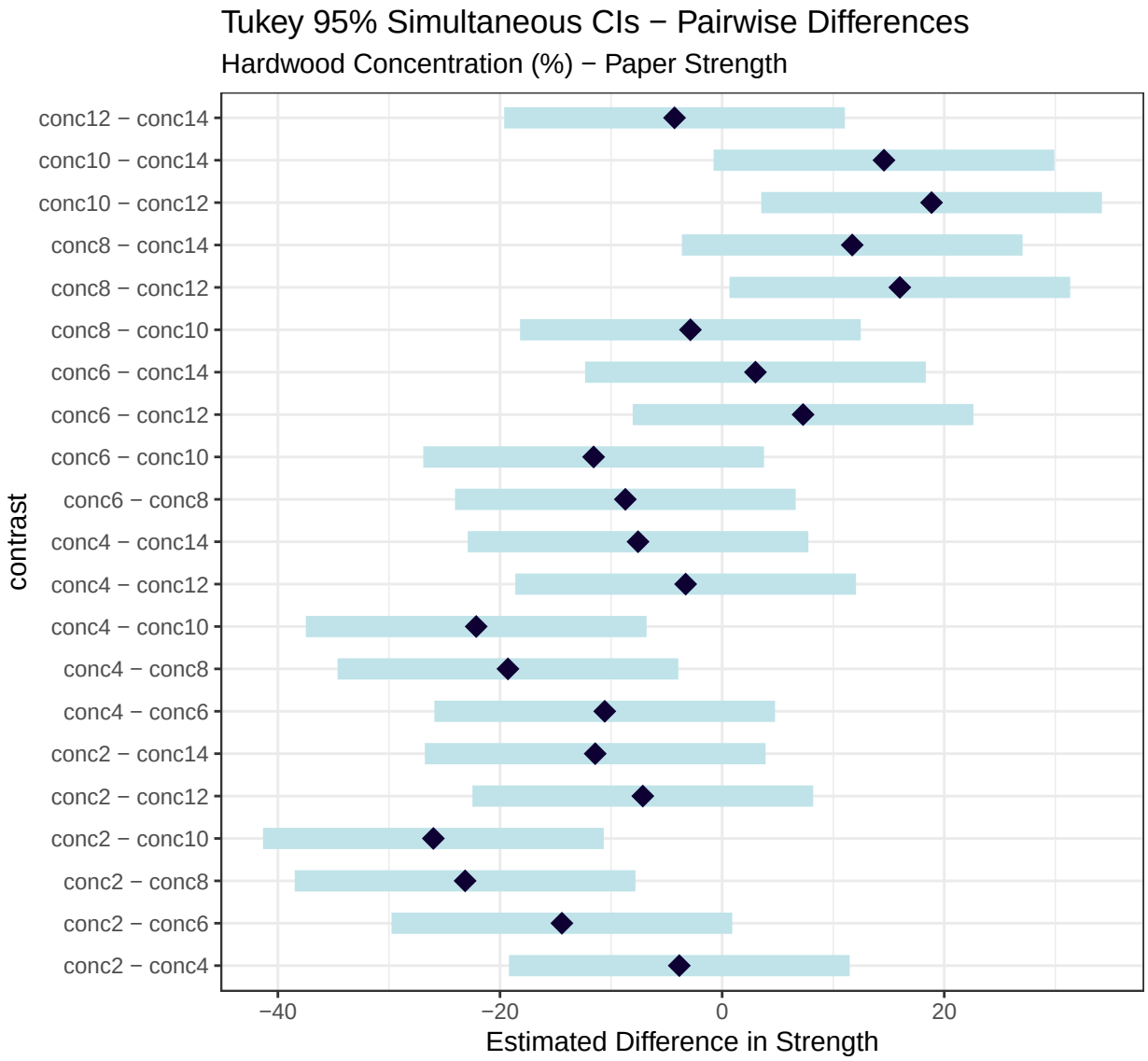
The LS means represent the adjusted treatment means, accounting for the imbalance in block structure. Higher hardwood concentrations tend to produce higher paper strength values, but formal multiple comparisons (Part e) will confirm which differences are significant.

Part (e): Tukey's Simultaneous 95% Confidence Intervals

The Tukey 95% simultaneous confidence intervals reveal which specific pairs of hardwood concentrations result in significantly different paper strengths. A pairwise difference is considered statistically significant if its confidence interval does not contain zero.

```
##  contrast        estimate   SE df lower.CL upper.CL
##  conc2 - conc4      -3.86 4.02  8  -19.196    11.48
##  conc2 - conc6     -14.43 4.02  8  -29.768     0.91
##  conc2 - conc8     -23.14 4.02  8  -38.482    -7.80
##  conc2 - conc10    -26.00 4.02  8  -41.339   -10.66
##  conc2 - conc12     -7.14 4.02  8  -22.482     8.20
##  conc2 - conc14    -11.43 4.02  8  -26.768     3.91
##  conc4 - conc6     -10.57 4.02  8  -25.910     4.77
##  conc4 - conc8     -19.29 4.02  8  -34.625    -3.95
##  conc4 - conc10    -22.14 4.02  8  -37.482    -6.80
##  conc4 - conc12     -3.29 4.02  8  -18.625    12.05
##  conc4 - conc14     -7.57 4.02  8  -22.910     7.77
##  conc6 - conc8      -8.71 4.02  8  -24.053     6.62
##  conc6 - conc10    -11.57 4.02  8  -26.910     3.77
##  conc6 - conc12      7.29 4.02  8   -8.053    22.62
##  conc6 - conc14      3.00 4.02  8  -12.339    18.34
##  conc8 - conc10     -2.86 4.02  8  -18.196    12.48
##  conc8 - conc12     16.00 4.02  8    0.661    31.34
##  conc8 - conc14     11.71 4.02  8   -3.625    27.05
##  conc10 - conc12    18.86 4.02  8    3.518    34.20
##  conc10 - conc14    14.57 4.02  8   -0.768    29.91
##  conc12 - conc14    -4.29 4.02  8  -19.625    11.05
##
```

```
## Results are averaged over the levels of: day
## Confidence level used: 0.95
## Conf-level adjustment: tukey method for comparing a family of 7 estimates
```



From our analysis we found:

- **Significant Differences:** Several concentration pairs show significant differences. For example, the intervals for `conc2 - conc8` ([-38.48, -7.80]) and `conc2 - conc10` ([-41.34, -10.66]) do not contain zero. This indicates that paper produced with 8% and 10% hardwood concentrations is significantly stronger than paper produced with a 2% concentration. Similarly, `conc4` is significantly weaker than both `conc8` and `conc10`.
- **The "Sweet Spot":** The concentrations of **8%** and **10%** consistently produce the highest paper strength. Looking at the `conc8 - conc10` comparison, the interval is [-18.20, 12.48]. Because this interval contains zero, there is no statistically significant difference in strength between 8% and 10% concentrations.
- **Higher isn't always better:** Pushing the concentration even higher to 12% or 14% actually causes strength to drop back down. For instance, `conc10` is significantly stronger than `conc12` (interval: [3.52, 34.20]).

Hardwood concentration significantly impacts paper strength. To maximize strength, the optimal concentration lies in the 8% to 10% range. Concentrations significantly lower (2%, 4%) or higher (12%) result in weaker paper.